

Supply Chain Analytics Project Report

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Tools Used: Excel, SQL (MySQL), Power BI

Role: Data Analyst

Project Type: End-to-End Analytics Project

1. Project Overview

- The objective of this project was to perform an end-to-end Supply Chain Analytics workflow starting from raw data cleaning to business insights and recommendations.
- The analysis focused on inventory optimization, supplier performance, and quality cost reduction using Excel, SQL, and Power BI.

2. Business Problem Statement

- The organization faced the following challenges:
- Frequent stock-outs affecting customer demand fulfilment
- Excess inventory increasing holding costs
- Supplier quality issues causing operational inefficiencies
- Lack of visibility into Cost of Poor Quality (COPQ)
- The goal was to analyze supply chain data and provide data-driven insights to support better decision-making.

3. Dataset Description

The dataset contained transactional supply chain data with the following key attributes:

- Product ID
- Supplier ID
- Supplier Name
- Demand Quantity
- Inventory On Hand
- Safety Stock
- Defect Quantity
- Safety stock
- Warranty cost
- Cycle cost etc

To maintain data integrity, the dataset was divided into:

Raw Data – Original unmodified dataset

Cleaned Data – Processed dataset used for analysis

4. Data Cleaning Process (Excel)

4.1 Raw Data Handling

- Raw data was stored separately and never modified
- Missing values and inconsistent formats were identified

4.2 Cleaning Steps Performed

- Removed duplicate records
- Standardized product and supplier names
- Converted date columns into proper date format
- Handled missing values using logical replacements

- Verified numeric columns for invalid values

4.3 Feature Engineering (Excel)

New calculated columns were created:

- Inventory Gap = Inventory On Hand – Safety Stock
- Stock Status (Stock-out / Safe)
- COPQ = Defect Quantity × Unit Cost
- The cleaned dataset was saved and used for SQL and Power BI analysis.

5. Exploratory Analysis Using Excel

- Pivot tables and charts were created to understand:
- Product-wise inventory levels
- Supplier-wise defect contribution
- Warehouse-wise inventory distribution
- This step helped in identifying initial trends and anomalies before deeper SQL analysis.

6. SQL Analysis (MySQL)

The cleaned dataset was imported into a MySQL database for structured analysis.

6.1 Key Business Questions Answered Using SQL

- Which products are below safety stock?
- Which products have excess inventory?
- Which suppliers contribute most to defects?
- What is the Cost of Poor Quality (COPQ) by supplier?
- How is inventory distributed across warehouses?

6.2 SQL Techniques Used

- WHERE conditions
- GROUP BY aggregations
- Calculated fields
- ORDER BY and filtering
- Business-driven query logic

Each query was saved as a separate .sql file for reusability and documentation.

7. Power BI Dashboard Development

7.1 Data Modelling

- Imported cleaned data into Power BI
- Created relationships between product, supplier, and warehouse data
- Implemented a star schema approach

7.2 DAX Measures Created

- Inventory Gap
- Stock-out Count
- COPQ
- Total Inventory
- Supplier Defect Contribution

7.3 Dashboard Pages

1. Dashboard Overview – High-level KPIs
2. Inventory Analysis – Stock-out and excess inventory insights
3. Supplier Performance – Defect contribution by suppliers
4. Quality & COPQ – Financial impact of poor quality
5. PPAP Analysis _ Quality Analysis

Slicers were added for:

- Product
- Supplier
- Month

8. Key Insights

1. Approximately 18% of products were below safety stock levels
2. One supplier contributed over 40% of total defects
3. Excess inventory was concentrated in specific warehouses
4. COPQ represented a significant hidden cost impacting profitability

9. Business Recommendations

Based on the analysis, the following actions were recommended:

1. Increase safety stock for high-demand products
2. Optimize inventory distribution across warehouses
3. Renegotiate contracts or improve monitoring for poor-performing suppliers
4. Implement quality control measures to reduce COPQ
5. Use dashboards for continuous supply chain monitoring

10. Project Outcome

This project successfully demonstrated:

1. End-to-end data analysis workflow
2. Strong understanding of supply chain KPIs
3. Practical use of Excel, SQL, and Power BI
4. Ability to convert raw data into actionable business insights

11. Conclusion

The Supply Chain Analytics project highlights how data-driven decision-making can improve operational efficiency, reduce costs, and enhance supplier performance.

This project reflects real-world analytical thinking and aligns with industry best practices for data analysis.

12. Tools & Technologies Used

- Excel – Data cleaning, KPI calculation, pivot analysis
- SQL (MySQL) – Business-driven querying and analysis
- Power BI – Interactive dashboards and visual storytelling

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